

Day 21: Confidence Intervals Code Automation: Manual and Library Methods

1. Approach 1: The Easy Manual Math Method (Step-by-Step Execution)

English: This method explicitly executes each algebraic component of the Confidence Interval formula. It avoids complex libraries and loop logic to keep the sequence of Standard Error, Margin of Error, and baseline shifts perfectly visible.

Hindi: Yeh approach pure Confidence Interval formula ko bina kisi loops ya advanced library ke line-by-line chalati hai. Isse aapko pata chalta hai ki kaise pehle Standard Error (SE) nikalta hai, phir Z-score se multiply hokar Margin of Error (MoE) banta hai, aur aakhir mein Mean me plus-minus hokar range lock hoti hai.

```
# =====
# METHOD 1: EKDUM EASY MANUAL CODE (BINA LOOP KE PURE MATH)
# =====

print("--- STEP 1: Given Input Data (Jewelry Case Study) ---")
sample_size = 100          # N = 100 orders
sample_mean = 25.0         # X = 25.0 grams (Sample Average)
std_deviation = 2.0        # σ = 2.0 grams (Asali Bhatkaw)
z_score = 1.96             # Z = 1.96 (95% Confidence Level ke liye fixed hai)

print("Sample Size (N) =", sample_size)
print("Sample Mean (X) =", sample_mean, "grams")
print("Standard Deviation (σ) =", std_deviation, "grams")
print("Fixed Z-Score for 95% =", z_score)

print("\n--- STEP 2: Standard Error (SE) Calculate Karna ---")
sqrt_n = 100 ** 0.5        # Root N = 10
standard_error = std_deviation / sqrt_n  # 2.0 / 10 = 0.2
print("Square Root of N =", sqrt_n)
print("Calculated Standard Error (SE) =", standard_error)

print("\n--- STEP 3: Margin of Error (MoE) Calculate Karna ---")
margin_of_error = z_score * standard_error  # 1.96 * 0.2 = 0.392
print("Calculated Margin of Error (MoE) =", margin_of_error, "grams")

print("\n--- STEP 4: Final Confidence Interval Range Build Karna ---")
lower_limit = sample_mean - margin_of_error  # 25.0 - 0.392 = 24.608
upper_limit = sample_mean + margin_of_error  # 25.0 + 0.392 = 25.392

print("Lower Limit (Mean - MoE) =", lower_limit, "grams")
print("Upper Limit (Mean + MoE) =", upper_limit, "grams")

print("\n--- FINAL EXECUTIVE INTERPRETATION ---")
print("Hum 95% poore bharose ke sath keh sakte hain ki hamari factory ke")
print(f"sabhi orders ka asli average weight {lower_limit:.2f} grams se")
print(f"{upper_limit:.2f} grams ke beech hai!")
```

2. Approach 2: The Industrial Library Method (Short & Efficient)

English: This approach uses automated modules from statistical libraries to abstract the execution of critical values and error spans into a optimized core block.

Hindi: Yeh approach statistical libraries ka use karke pooray calculations ko abstract kar deti hai. Isme aapko na toh manual standard error nikalna hai na hi plus-minus karna hai, library backend par Z-score ko automated tareeqay se control kar ke direct ranges de deti hai.

```
# =====  
# METHOD 2: PROFESSIONAL INDUSTRIAL LIBRARY METHOD (SHORT VERSION)  
# =====  
  
from scipy import stats  
  
# STEP 1: Given Data Input Karo  
n, mean, sd = 100, 25.0, 2.0  
  
# STEP 2: Single Line Me Range Nikalo (Confidence=0.95, Loc=Mean, Scale=SE)  
lower_limit, upper_limit = stats.norm.interval(0.95, loc=mean, scale=sd / (n **  
0.5))  
  
# STEP 3: Result Print Karo  
print(f"95% Confidence Interval Range: [{lower_limit:.2f}, {upper_limit:.2f}]  
grams")
```